

Performance Testing

Date	3 July 2026
Team ID	XXXXXX
Project Name	AI-Crop recommendation System
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how the OptiCrop application performs under different user loads and verifies that the Machine Learning crop recommendation system provides fast, accurate, and reliable crop predictions based on soil nutrients and environmental conditions

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing
Target Module / API	AI crop recommendation, Flask Prediction Route (/submit), Loan Approval Prediction
Test Environment	Local System (Windows 11)
Test Date	02-07-2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Home Page Loading	20	60	Home page should load successfully with minimal response times.
2	Crop Recommendation Form Submission	50	120	Soil and environmental data should be submitted success.
3	Crop Recommendation Prediction	100	180	Recommended crop should be generated accurately without server failures

4	Multiple Users Accessing the Application	75	120	Application should remain responsive under concurrent user requests.
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Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.4 seconds	pass	Response time within acceptable limits.
2	Response Time (Max)	< 5 seconds	3.6 seconds	Pass	Stable during peak load.
3	Throughput (Req/sec)	>40 Req/sec	50 Req/sec	Pass	Good request handling
4	Error Rate	< 1%	0.1%	Pass	Very few request failures
5	CPU Utilization	< 80%	63%	Pass	CPU usage remained stable
6	Memory Utilization	< 80%	56%	pass	Memory usage within limits.

Step 4: Observations & Analysis z

Key Findings:	• The Flask backend processed crop recommendation requests efficiently with low response time. • The Machine Learning model accurately recommended suitable crops without any server failures during testing.
Bottlenecks Identified:	• A slight increase in response time was observed when multiple users accessed the application simultaneously. • Processing large input requests under heavy load may slightly increase crop prediction response time.
Optimization Steps Taken:	• Optimized the Flask application by efficiently preprocessing soil and environmental input before prediction. • Loaded the trained Machine Learning model and StandardScaler at application startup and improved input validation for faster, reliable crop recommendations.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):



